Learn and Live Opportunity Index: Mapping Housing & Education Inequality Across Metropolitan Richmond, Virginia

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Introduction

- **Living conditions and neighborhood environments** directly influence children's **educational attainment, health outcomes, and long-term socioeconomic mobility** (Chetty et al., 2018; Acevedo-Garcia et al., 2024).
- Many indices have been created to describe patterns of inequality. Indices make it easier to show how **multiple variables intersect** in order to create a more **comprehensive view** of **patterns of inequality**.
  - Diversitydatakids.org created the **Child Opportunity Index** (COI) using 44 indicators across education, environment, and socioeconomic areas by census tract and block group (Acevedo-Garcia et al., 2024).
- This project builds off of research and work provided in the **2017 Live and Learn Report**, and the **2024 Live and Learn 2.0 Report**. Findings demonstrated historical patterns and policy suggestions around **housing and education-based segregation** in the metro Richmond area.
  - The **Live and Learn Opportunity Index (LLOI)** uses **demographic, education, and housing** variables to determine the **distribution of opportunity** across school zones.
  - School attendance zones are **unique policy defined geographic boundaries** that determine where students are assigned their education, acting as a **unique spatial unit for the LLOI**, and framework for examining patterns of inequality (Owens, 2018; Monarrez et al., 2022).

Research Questions

1. How can educational, housing, and neighborhood/demographic conditions be constructed into a single measure of opportunity by school attendance zones?
2. How does opportunity vary by attendance zone within metropolitan Richmond?
3. How does opportunity intersect with race and affordable housing?

Methodology and Variables

- **Selected variables** for the LLOI based on **established frameworks** and **Live & Learn research** (Acevedo-Garcia et al., 2024).
- **Estimated census-based data to attendance zones** using areal interpolation (areal package 0.1.8) and standardized variables using z-scores in R.
- **Compared the LLOI** to additional **demographic & housing variables**, and ran correlations of the LLOI against other opportunity indices.
- **Mapped and analyzed** spatial patterns of opportunity across school zones using ArcGIS Pro.

$$LLOI_j = P\left(\sum_{i=1}^{14} d_i \cdot \frac{x_{ij} - \bar{x_i}}{s_i}\right)$$

P = Percentile Rank of composite scores
d_i = direction coefficient orienting values to same direction
x_{ij} = raw value of variable *i* in zone *j*
x_i = mean of variable *i* across all zones
s_i = standard deviation of variable *i* across all zones

Figure 1. This shows the formula for the summed z-scores which were used to create the LLOI.

Variables for the LLOI were chosen based off the **Live & Learn 2.0 Report**, that have direct **relationships to opportunity, public availability, relevancy, and balancing**

Name	Category	Source	Date
Median Household Income	Demographic	American Community Survey(ACS)	2024
Bachelor's Degree	Demographic	ACS	2024
Non-English Speaking Households	Demographic	ACS	2024
Households Receiving SNAP/Food Stamps/Income Assistance	Demographic	ACS	2024
<18 Without Health Insurance	Demographic	ACS	2024-25
Economically Disadvantaged Students	Education	Virginia Department of Education(VDOE)	2024-25
Chronic Absenteeism	Education	VDOE	2024-25
Provisionally Licensed Teachers	Education	VDOE	2024-25
Reading Proficiency Pass Rates	Education	VDOE	2024-25
Housing Cost Burden	Housing	Department of Housing and Urban Development (HUD)	2025
Small Area Fair Market Rent	Housing	HUD	2025
Neighborhood Poverty Rate	Housing	ACS	2024
Eviction Filing Rate	Housing	RVA Eviction Lab	2018-25

Table 1. Each variable falls under a foci of (1) demographic (2) education, or (3) housing, recorded between 2024-25 from public agencies.

Mapping the LLOI by Attendance Zone: High Schools

The LLOI measures opportunity across **134 elementary, 40 middle, and 32 high school attendance zones** across the eight jurisdictions.

- High schools across the eight counties illustrate **broad concentrations and patterns of opportunity** across the are

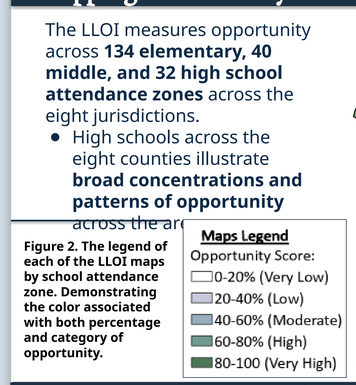


Figure 2. The legend of each of the LLOI maps by school attendance zone. Demonstrating the color associated with both percentage and category of opportunity.

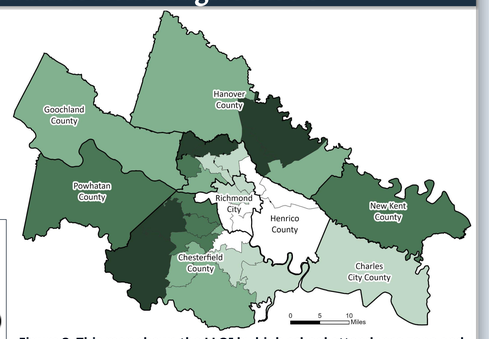


Figure 3. This map shows the LLOI by high school attendance zone and jurisdiction. Thicker black lines indicate county borders and thinner grey lines indicate school boundaries.

Middle Schools

Many regional opportunity patterns **are consistent** at the middle school level.

- The urban core of the metro area in Richmond city and Henrico county show **concentrations of low opportunity**.
- Areas of **high opportunity** can be located across **more rural areas** of Chesterfield, and Hanover counties.
  - **denser geographic areas** and school zones **exhibit overall less**

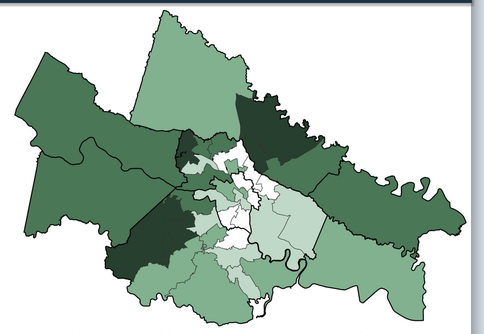


Figure 4. The LLOI by middle school attendance zone. Thicker black lines indicate county borders and thinner grey lines indicate school boundaries.

Elementary Schools

Elementary attendance zones **reveal the greatest variation in opportunity.**

- **Higher-opportunity** is clustered in western Henrico, Hanover, and portions of Chesterfield.
- **Lower-opportunity zones** are concentrated in Richmond City and eastern portions of Henrico.

These spatial patterns **are not constrained by county or city boundaries.**

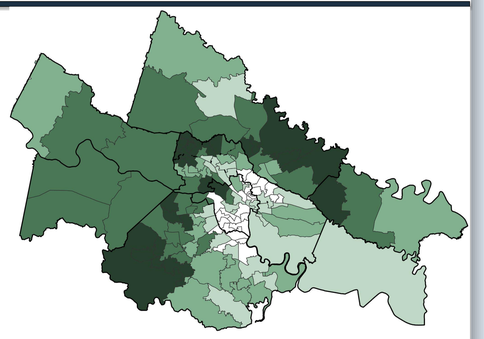


Figure 5. The LLOI by elementary school attendance zone indicates more geographic variation in opportunity. Thicker black lines indicate county borders and thinner grey lines indicate school boundaries.

Mapping the LLOI & Affordable Housing Units

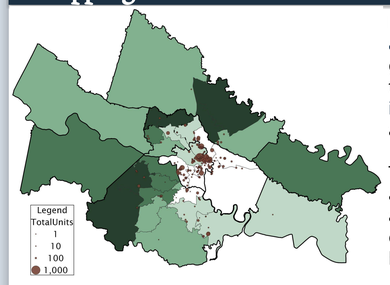


Figure 6. Total affordable housing units are represented as proportional symbols over high school opportunity zones. The highest amount of LIHTC Units recorded in the dataset was 781 in Richmond City.

Low-Income Housing Tax Credit (LIHTC) is an **affordable housing program** where the federal government offers tax credits to states, who give them to private developers in order to **incentivize** the creation and distribution of affordable housing (Keightley, Mark 2025).

This map provides **visualization of** LIHTC affordable housing units across high school attendance zones. Two patterns emerge between concentrations opportunity and subsidized housing:

- **LIHTC units are most concentrated in the city center** and **densely populated urban areas**
- **LIHTC units are concentrated within** high school boundaries **with lower opportunity.**

LLOI and Race

The LLOI considers the **characteristics** that **impact individuals who live in a given area**. The presence of race and racial segregation is important to the LLOI, often demonstrated as consequences to the **structures of inequality** found in **exclusionary housing and education practices** (Monarrez, T., Choi, C., LaFortune, J., 2022). Which, is why we chose to **compare race against the LLOI** in a **post hoc analysis** rather than in the index itself.

To further examine this relationship, we compared the percentage of Black residents within each elementary attendance zone with its LLOI score.

- Elementary attendance zones with **higher concentrations** of Black individuals generally had **lower opportunity scores.**
- The R^2 value indicates that **65% of the variation in opportunity scores** can be explained by **change in racial composition** across attendance zones.
- This pattern is **consistent with research** showing that attendance boundaries **often reflect residential segregation** and access to opportunity (Owens, 2018; Monarrez et al, 2022).

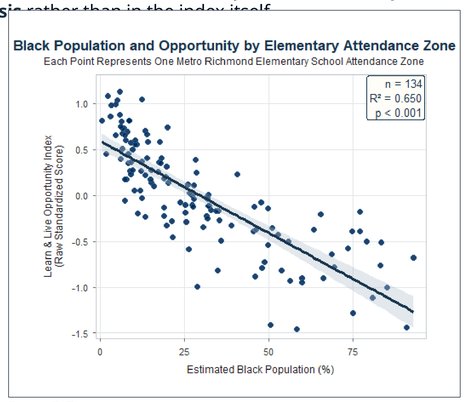


Figure 7. This regression shows the relationship between the percent black population by school attendance zone with the LLOI standardized scores.

Testing for Correlation: COI vs. LLOI

The LLOI is **highly correlated** with opportunity values demonstrated through the COI, however the **methods and indicators** of each index **differ**.

- The COI is more **comprehensive in scoring opportunity** by including variables related to factors like health, safety, pollution, and employment.
- The LLOI is unique in the foci of **education and housing data**.
  - The LLOI includes **additional housing variables** not demonstrated in the COI, and mapped over a **regionally unique spatial** unit of analysis.

Attendance Zone	Mean Difference (LLOI - COI)	P-value	Spearman's P
Elementary	-0.1958	< 0.001	0.9393559
Middle	-0.2761	< 0.001	0.9221388
High	-0.3198	< 0.001	0.9109238

Table 2. Results of Pearson's correlation between Child Opportunity Index and the Live and Learn Opportunity Index are demonstrated above. Results show that the indices are highly correlated.

Discussion

- The LLOI maintains patterns with **[illegible]** of variables in the **Live and Learn 2.0 Report** mapped by **school attendance zones** rather than census tracts.
- School attendance zones are **policy-relevant geographic units**, useful for understanding opportunity through boundaries that determine **access** to public education. They provide a **finer spatial resolution to reveal localized variation in opportunity**, especially at the elementary level.
- The LLOI and COI are very similar, **supporting the validity of the LLOI** while demonstrating the value of incorporating **region-specific** housing and educational indicators.
- **Strong correlation** between race and the LLOI reflect **historical trends** of housing and race-based segregation, and **modern discussions** of school zoning it's impact on educational opportunity (Owens, 2017; Monarrez et al., 2022; Castro, A., et al., 2024).

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Future Work

The LLOI establishes a **locally relevant measure of opportunity** that can be replicated through R as new data becomes available.

Policymakers and local legislators can use the LLOI to **identify schools that need the most investment**, and **where to equitably distribute local support initiatives** throughout the region.

References